

NEAR-REAL-TIME SCREENING OF MICROBIAL CONTAMINATION IN FOODS

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INTRODUCTION

Microbial contamination remains a major challenge for the food industry, contributing to food spoilage, foodborne illness outbreaks, product recalls, and substantial economic losses. Timely identification of microbial contamination is therefore critical for maintaining food safety and quality. However, conventional culture-based methods typically require 2–4 days to obtain preliminary results and longer for confirmation. In industrial settings, samples are also frequently transported to off-site laboratories for analysis, introducing further delays that can hinder timely decision-making and reduce product shelf life. Consequently, there is a need for rapid, on-site screening technologies capable of providing early indications of microbial contamination. Microbial volatile organic compounds (mVOCs), which are generated during microbial growth, have potential as indicators of contamination. This study therefore investigated whether mVOCs can be used as early indicators of microbial contamination in food and whether mVOC-based approaches could complement conventional microbiological methods by enabling fast, on-site screening.

METHODS

UHT milk and commercial mixed leafy salad were selected to represent a microbiologically clean matrix and a complex matrix with naturally occurring background microflora, respectively. Food samples were inoculated with an *E. coli* cocktail comprising ATCC 25922, ATCC 8739 and K-12. UHT milk was inoculated at 1 CFU/mL, while mixed leafy salad was inoculated at 100 CFU/g, followed by incubation at 37 °C for 24 h. Headspace mVOCs were profiled using headspace solid-phase microextraction coupled with gas chromatography–time-of-flight mass spectrometry (HS-SPME-GC–TOF-MS), whereas GC×GC–TOF-MS was employed to enhance chromatographic separation and resolution, enabling the detection of volatile compounds that may have been unresolved or missed by one-dimensional GC. Putative mVOCs associated with *E. coli* contamination were identified by comparison with mass spectral databases and reference standards. Culture-based microbiological analysis was performed alongside volatile profiling to confirm the microbial counts.

RESULTS AND DISCUSSION

In UHT milk, HS-SPME-GC–TOF-MS identified multiple mVOCs associated with *E. coli* growth. Ethanol and acetic acid were prominent in *E. coli*-spiked samples and were significantly different from controls ($p < 0.05$). In mixed leafy salad, indole was prominent in *E. coli*-spiked samples and significantly differentiated spiked samples from controls ($p < 0.05$), despite the complex food matrix and high background microflora. These findings demonstrate that informative mVOC signals can be detected in both a low-background matrix (UHT milk) and a complex matrix (leafy salad) (**Figure 1**).

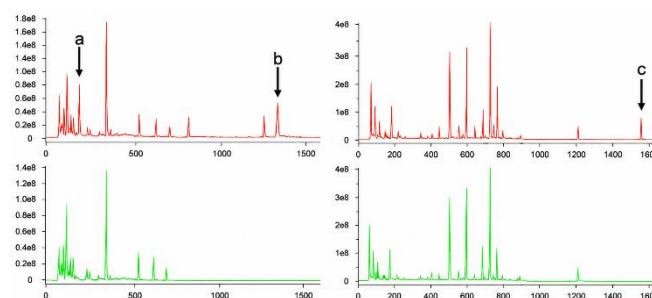


Figure 1. Early indication of *E. coli* contamination in model food matrices based on mVOC profiles. (A) mVOC profiles of UHT milk spiked with an *E. coli* cocktail at an initial inoculation level of 1 CFU/mL and incubated at 37 °C for 24 h, showing the detection of (a) ethanol and (b) acetic acid in the spiked samples. (B) mVOC profiles of mixed leafy salad spiked with the same *E. coli* cocktail at an initial inoculation level of 100 CFU/g and incubated at 37 °C for 24 h, showing the detection of (c) indole in the spiked samples. Orange and green chromatograms represent spiked and control samples, respectively.

CONCLUSIONS

This study demonstrates the potential of mVOC profiling for the early screening of microbial contamination across food matrices with different levels of complexity. Distinct mVOCs associated with *E. coli* contamination were identified in both UHT milk and mixed leafy salad, despite differences in their background microflora. These findings support the potential use of mVOCs as indicators of microbial contamination and provide a basis for developing rapid, on-site sensing technologies. Such approaches could complement conventional microbiological methods by enabling earlier screening and supporting timely decision-making in food production environments.